

STAT 190: Introduction to Statistical Methodology

Spring 2026

Instructor:	Dr. Aruni Jayathilaka. Statistics Program Hylan 703 University of Rochester aruni.jayathilaka@rochester.edu
Office Hours:	Please refer to Blackboard for the schedule.
Course Details:	4.0 Credits TR 2 – 3.15 p.m. Hutchison 141
Prerequisites:	MATH 141, or equivalent
Course Description:	This course is an introduction to statistical methodology, focusing on the probability and statistical theory underlying the estimation of parameters and testing of hypotheses. Students are exposed to basic data exploration, summarization of graphical display of data, axioms of probability, distributions and related theory, parameter estimation, and statistical inference. Advanced topics include linear correlation and regression analysis. Students will perform calculations with statistical software such as R/RStudio.
Credit Hours:	This course follows the University's credit hour policy for four-credit courses. Four credit-hour courses include 150 minutes per week of lecture time (two lectures at 75 minutes each), 50 minutes of independent study and lab work, and 450 minutes per week of out-of-class student work (e.g., reading, assignments, study time, and other unsupervised work).
Technology:	Students will need a laptop or computer with at least 8GB of memory, 256GB of local storage and reliable internet access. (Only in Labs) A calculator (basic, scientific, or graphing) will be needed for exams. It should have basic functions including the square root function and the mathematical constant, e. Calculator apps on cell phones, laptops or tablets are NOT permitted on exams. We will not cover graphic calculator commands in class.
Software:	R, RStudio, and RMarkdown will be needed for this course. Information on accessing such software is available on the course Blackboard page.

Course Objectives & Learning Outcomes:

This course aims to provide students with basic statistical competencies, focusing on theory, application, and computational methods for analyzing data. By the end of this course, students will be able to:

- Appropriately summarize and display data
- Describe relationships between multiple variables.
- Apply probability axioms to random variables
- Determine and apply the appropriate statistical test for a hypothesis
- Build and interpret regression models
- Analyze data using R, both through built-in and custom functions
- Compile results into cohesive reports using RMarkdown.

Course Policies & Expectations:

It is expected that students will come to class with an open mind for learning statistics. Statistics should be learned in an environment that mirrors how it will be used in application: in a collaborative manner. Thus, I will promote discussion of what we are learning in class. Students are highly encouraged to participate in class by asking questions, noting confusions, and providing examples from their other courses. While this class is set up to have traditional lectures, it is expected that those lectures are forums for discussion between students and the instructor. For this reason, attendance is important. While I understand other obligations may arise that preclude you from making it to a class, please make every effort to attend every class.

Laptops or tablets are NOT permitted in class EXCEPT in tablet mode for handwritten notes OR if you have accommodation from Disability Resources. Use of these devices for any other non-educational reason is not permitted. Please silence all electronic devices. Please limit all side conversations during class time.

Materials & Access: There is no required text for this course. Course notes and assignments will be placed on Blackboard.

- Course notes:
 - Student slides will be posted at the beginning of the week. These slides can be downloaded, and you can bring printouts or use a tablet to complete notes.
 - Examples in the slides are not complete and will be discussed during class and so can be filled in during the lecture. Therefore, you should prioritize attending lectures and paying attention to the lecture.

Assignments & Grading Procedure:

Grading and assessment for the course will be broken into the following pieces:

1. Homework Assignments: 30%

There will be ~ approximately 6 homework assignments. Assignments will be posted to Blackboard, and students are expected to submit their assignments through Blackboard as well. Homework will generally be assigned on a biweekly basis and due dates for assignments will be clearly written on each assignment. Generally, homework assignments will involve the use of R and RMarkdown to analyze datasets from a computational perspective. Late work will not be accepted unless arrangements are made well in advance or in extenuating circumstances with valid documentation.

Statistics is a collaborative field, and you are encouraged to discuss with other students, but you are responsible for your own assignment. Homework assignments are allowed to be verbally discussed between students. Students are **not** permitted to share code, output, or written summaries with each other. The work you submit must be your own, and each student must complete and submit their own assignment. Please see the **Academic Integrity** section of the syllabus for more information.

2. R Labs – 20% Total

Students must attend a near-weekly R Lab to learn how to code statistical methods in R!

20% = 10% for attendance + 5% complete assignments + 5% for correctness of lab assignment

The weekly lab assignments will be based on attendance, completion of the lab tasks and questions and correctness – I'm not expecting 100% correctness, it's okay to make mistakes in programming! You may miss TWO (2) labs without penalty.

R Labs begin the 2nd week (Sep 1st week) of the semester on either a Tuesday, Wednesday or Thursday based on the day you registered for in the location listed. There is no Zoom/virtual option.

If you have a temporary conflict with your lab time and need to attend an alternate time slot for that week **ONLY**. (This is an emergency protocol. You cannot use this option more than three times.)

Temporary Switches Protocol:

- You are required to attend your assigned lab that you registered for unless you need a temporary switch.
- You need to email the current lab TA and TA for the session you wish to attend with the request and **CC the lab head TA** too. (Lab head TA will just observe the situation and will not reply unless necessary)
- The TA for the session you wish to attend can approve or deny the request based on student request frequency and room cap. Labs should not exceed 20 students in any given week.
- If the session you wish to attend is over cap you will need to go with an alternative lab TA.
- Although you switched the session you need to email your work to your assigned TA.

If you enrolled in this course on the last day of the add/drop period, there is a possibility you will miss Lab 1. It is your responsibility to contact your instructor/TAs to keep up to date with materials. If you don't make lab 1 you will have only one lab remaining to miss without a penalty.

3. Midterm Exam: 20%

There will be one midterm exam, worth 20% of your total grade. These will take place at your lecture hall in the regular lecture time. The midterm exam is not open to any collaboration and will be held in-person. Students will complete the exam independently.

Near the time of the exam, we will go over the expected test material in class. Formula sheets will be provided. Students are only permitted to use calculators for additional aid.

4. Final Exam: 30%

There will be a cumulative final exam, worth 30% of your total grade. You will be informed of the time and location of the exam near the exam date. Near the time of the exam, we will go over the expected test material in class. Formula sheets will be provided. Students are only permitted to use calculators for additional aid.

5. Bonus/Reward points:

- There are no extra credits to replace your homework/ labs/ exams.
- There will be some bonus questions for some homework and exams.
- There will be a reward point system to encourage you to attend lectures, attend labs, submit your homework assignments and attend exams on time. (Details will be discussed on the 1st day and you can find details in Blackboard.)
- There is **NO** make-up for bonus/reward points.

There will be **NO** make-up exams unless in extenuating circumstances with valid documentation. Please refer to the course schedule below for dates of exams. Plan accordingly for end of semester travel.

The lecture hall has the ability to record our lectures for emergency situations (if you have an emergency situation, please notify me ASAP and provide appropriate documentation). There is no guarantee of good video quality – this is NOT a replacement for live lectures.

Letter grades will be assigned as follows after rounding your final grade to the nearest integer (e.g., 93.5 rounds up to an A, whereas 93.4 rounds to an A-):

A: 94-100, A-: 90-93, B+: 87-89, B: 83-86, B-: 80-82, C+: 77-79, C: 73-76, C-: 70-72, D+: 67-69, D: 63-66, D-: 60-62, E: 0-59.

Teaching Assistants & R Lab Leaders:

Class TAs: You can meet any class TA during their office hours, but you are only permitted to email your assigned class TA.

Your last name begins with:	Class TA	Email
A-I	Haohan Ying	hying4@u.rochester.edu
J-R	Amanda Jen	ajen@u.rochester.edu
S-Z	Madeline Jen	mjen@u.rochester.edu

The class TAs will grade homework assignments and be a resource for you during the course. You are assigned to your TA as above and your TA will be your primary contact person throughout the course.

Lab Leaders:

Lab TA	Sections	Email
Mai Pham (Head lab TA)	9 only	mpham8@u.rochester.edu
Yu-Tang Wang	3 & 5	ywang434@u.rochester.edu
Adam Goldschmidt	6 & 10	agoldsc2@u.rochester.edu
Yi Zhou	11 & 15	yzhou113@u.rochester.edu

Generally, your lab TA will be grading your lab assignments and should be the person you email with lab assignment and general Rstudio/RMarkdown questions.

There are a lot of students and only a few TAs, so having everyone stick to their assigned class and lab TAs allows them to provide you with more efficient answers and support. When you email a TA, you will get a response within 24 hours (keep that in mind if you are waiting until the last minute to start your assignments...).

Academic Honesty: Academic honesty is a core value of the University of Rochester. Students who violate the [University of Rochester University Policy on Academic Honesty](#) are subject to disciplinary penalties, including the possibility of failure in the course and/or dismissal from the University. Since violations of academic honesty harms the individual, other students, and the integrity of the University, policies on academic honesty are strictly enforced.

The work you submit is expected to be your own and an honest representation of your knowledge and efforts. Online solution sets and paid online resources (e.g., chegg.com or similar) are not permitted in any way for this course and usage of these is considered a violation of the Academic Honesty policy. Similarly, generative AI (e.g., ChatGPT or similar) is not permitted to be used when completing homework assignments and labs. These applies even if the unauthorized content was obtained through indirect means (e.g., through a friend) and/or the student is seemingly unaware that that content originated from such sources.

Generative AI is permitted to be used, though, for general knowledge and learning. If you want to use generative AI as a resource to help in your general understanding of topics, that is permitted. The key word here is *general*. You should only be using generative AI to help with overarching subject knowledge and not specific tasks. To make this distinction clear, let me present an example. Suppose you have a homework assignment question asking you to conduct an Anova test on a set of data and interpret it. You *can* use generative AI to learn more about Anova tests generally, their overall uses, and broad interpretation. You *cannot* ask it to help conduct the test on your particular dataset, provide code for completing this task, nor interpret the results in the particular context we are looking at. You essentially should only use generative AI if you are going through the notes and want additional explanations on broad concepts.

Please refer to the following for assignment specific academic honesty policies:

- Labs and Homework Assignments: Statistics is a collaborative field, and you are encouraged to verbally discuss assignments with other students, but you are responsible for your own assignment. Students are not permitted to share code, output,

screenshots, or written summaries with each other—only verbal discussions of assignments are permitted between students in the course. Each student must submit their own assignment that is a reflection of the work they themselves completed.

- Exams: The exams are not open to any collaborations. Students will complete the exams independently. Any permitted resources that may be used when completing the exams will be clearly communicated by the professor in advance of the exam.

Accommodation: Students needing academic adjustments or accommodations because of a documented disability must contact the Disability Resource Coordinator for the school in which they are enrolled. I am happy to accommodate all accommodations, so long as they are documented with the Office of Disability Resources. I am glad to meet to discuss your specific situation or to help ensure you have the support you need. For additional information, please see: <https://www.rochester.edu/college/disability/>.

Electronic Communication Policy:

Announcements for the whole class will be sent via Blackboard Announcements. If you believe you are not receiving these announcements, check our Blackboard course and contact IT. If your question/concern is private, please schedule a one-on-one meeting with me or you can email.

If you have a course question, please try these methods *in this order*:

1. Check if your question was answered in the notes and examples we did in the class.
2. Come to instructor's office hours.
3. Email or meet TAs in their office hours.
4. Email the questions. (Please mention your class and section when emailing me)

Setting boundaries on electronic communication is healthy. I typically check my email and respond on weekdays (Monday – Friday) between 9am - 6pm. I don't check my emails frequently on the weekends. PLEASE remember this when completing homework assignments; if you are completing your assignments last-minute over the weekend and you run into a problem, it is likely that I will not be immediately available to assist you.

If you need immediate help, here are some great resources:

- Public Safety, **Call or text (585) 275-3333**, <https://www.rochester.edu/public-safety/>
- Counseling Center, (585) 275-3113, <http://www.rochester.edu/uhs/ucc/>
- University Health Service, <http://www.rochester.edu/uhs/>
- Interfaith Chapel, <https://www.rochester.edu/chapel/>

Title IX:

All members of the UR community have the right to learn and work in a safe environment free from all forms of discrimination, harassment, and related retaliation. This includes discrimination and harassment based on protected characteristics, and Title IX-prohibited behaviors, which include sexual assault, dating and domestic violence, and stalking. Students who have been subjected to sexual misconduct have the right to receive academic and [other supportive measures](#), to receive counseling and health services, and to make a report to the university and/or to law enforcement. In addition, [students who are pregnant, have recently experienced childbirth, and/or have medical needs related to childbirth](#) can

contact the Title IX office for assistance. All [faculty and teaching assistants are mandatory reporters](#) who must report Title IX-prohibited behavior to the Title IX office.

Hardship Policy: If you are in a situation where you are unable to participate in your courses as you planned, please contact CAS for advice on short-term accommodation. This typically applies to situations like deaths in the family, hospitalization and natural disasters. CAS will be able to notify all of your instructors of your hardship should it qualify. Long-term absences may be better suited with leaves of absence or incompletes. Incomplete for this course will be considered for students who have completed at least 75% of the coursework and their completion was compromised due to documented circumstances out of their control. Students requesting an incomplete should do so via email BEFORE the last day of class.

Incomplete Policy: It is expected that you are making this course a priority and putting in the effort to stay on the assigned course timeline. The incomplete notation will only be an option in the case of a documented circumstance beyond your control that keeps you from reasonably being able to finish the course. [School policy](#) requires dean approval for the incomplete notation, and 1) there must be documented evidence of a circumstance beyond your control keeping you from completing the course on time, 2) the majority of the course work must already be complete, and 3) only involve the work described in the syllabus.

Course Schedule: Below is a **tentative** course schedule. This is subject to change based on specific class interests and needs. Please always refer to Blackboard for the most updated copy of the schedule.

Week	Class dates	Topic	Labs
Jan 19 - 25	Jan 20	Syllabus and Introduction to statistics (CH1)	R Lab 0 - Introduction to Rstudio & RMarkdown (These lab materials are self-learning)
	Jan 22	Descriptive statistics (CH 2)	
Jan 26 – Feb 1	Jan 27	Descriptive statistics (CH 2)	R Lab 1 – Descriptive statistics
	Jan 29	Relationships (CH 3)	
Feb 2 - 8	Feb 3	Relationships (CH3) + Probability (CH4)	R Lab 2 – Relationships between variables
	Feb 5	Probability (CH 4)	
Feb 9 - 15	Feb 10	Probability (CH 4)	R Lab 3 – Probability problems.
	Feb 12	Probability (CH 4)	
Feb 16 - 22	Feb 17	Probability (CH4)	R Lab 4 - Probability problems.
	Feb 19	Distributions (CH5)	
Feb 23 – Mar 1	Feb 24	Distributions (CH5)	Mid exam review (Attendance is recommended for exam 1 bonus points.)
	Feb 26	Distributions (CH5)	
Mar 2 - 8	Mar 3	Distributions (CH5)	No Labs
	Mar 5	Mid Exam	
Mar 9 -15	Mar 10	Spring Break (No Classes)	Spring Break – No Labs
	Mar 12		
Mar 16 -22	Mar 17	Confidence intervals (CH6)	Lab 5 – Distributions
	Mar 19	Confidence intervals (CH6) + Hypothesis testing for one sample (CH7)	
Mar 23 - 29	Mar 24	Hypothesis testing for one sample (CH7)	R Lab 6 – CI & one sample HT
	Mar 26	Hypothesis testing for one sample (CH7)	
Mar 30 – Apr 5	Mar 31	Hypothesis testing for two sample (CH8)	Inference Practice (Attendance is recommended for homework 4 bonus points.)
	Apr 2	HT for 2 sample (CH8) + Variances (CH9)	
Apr 6 - 12	Apr 7	ANOVA (CH10)	R Lab 7 – Two sample HT & variances
	Apr 9	ANOVA (CH10)	
Apr 13 - 19	Apr 14	Inference of proportions (CH 11)	R Lab 8 – ANOVA
	Apr 16	Inference of proportions (CH11)	
Apr 20 - 26	Apr 21	Chi-square tests (CH 12)	R Lab 9–Proportion
	Apr 23	Regression (CH 13)	
Apr 27 – May 3	Apr 28	Regression (CH 13)	R Lab 10 – Chi-square
	Apr 30	Regression (CH 13) + Final exam review	R Lab 11 – Regression
May 4 - 10		Exam week	Saturday, May 9, 4:00 p.m.

